ाई आर – राष्ट्रीय भौतिक प्रयोगशाला
ATIONAL PHYSICAL LABORATORY
(वैज्ञानिक तथा औद्योगिक अनुसंधान परिषद्)
(Council of Scientific and Industrial Research)
ष मापिकी संस्थान (एनएमआई), सदस्य बीआईपीएम एवं हस्ताक्षरकर्ता सीआईपीएम - एमआरए)
ional Metrology Institute (NMI), Member BIPM and Signatory CIPM - MRA)
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अंशांकन प्रमाण पत्र
CALIBRATION CERTIFICATE
A SET OF WEIGHTS

प्रमाण पत्र संख्या / Certificate No.

25060361/D1.01/C-051

दिनांक/ Date	अगले अंशांकन हेतु अनुशंसित तिथि Recommended date for next calibration	पृष्ठ / Page	पृष्ठों की संख्या / No. of Pages
22.08.2025	22.08.2028	1	2

1. **Calibrated for** : National Test House (SR)
Department of Consumer Affairs
CSIR Road, Tharamani
Chennai - 600113, Tamilnadu, India

Customer's Ref. No. : NTH/17(1)/2025-26 Dated : 29.04.2025

2. **Description and identification of instrument**

10 kg to 1kg (05 Nos.)	:	Integral Knob cylindrical stainless steel weights.
500 g to 1 g (12 Nos.)	:	Cylindrical stainless steel weights.
500 mg to 1 mg (12 Nos.)	:	Stainless steel wire weights
Assumed Density (d)	:	(7950 ± 140) kg/m^3; $k = 2$
Make	:	Weigh India

3. **Environmental conditions** : Temperature : (23.0 ± 1.5) °C
Relative humidity : (50 ± 10) %
[Change in temperature during the calibration was less than ± 0.7 °C per hour]

4. **Standard(s) used with associated Measurement uncertainty** : NPL working standard (Sub-Div.#1.01/M/224(2)/00 & Sub-Div.#1.01/M/60027/55) of mass with uncertainty better than one-third of the reported uncertainty of measurement.

5. **Metrological Traceability of Standard(s) used** : The standard(s) used for calibration is(are) traceable to the National Standard which realize the units of quantities according to the International System of Units (SI).

6. **Principle/Methodology of Calibration & Calibration Procedure number** : CSIR-NPL Calibration Procedure No. Sub-Div.#1.01/Doc.3/ CP#WT/M-03. Method of comparison with the CSIR-NPL working standard(s) using substitution weighing. The reported mass value(s) is(are) the conventional mass value(s) (M_c) related to the true mass value(s) (M_t) by formula : $M_c = M_t \times [1 - 1.2 \times (1/d - 1/8\,000)]$.

अंशांकनकर्ता:
Calibrated by:
(D.C. SHARMA)

जाँचकर्ता:
Checked by:
(NIDHI SINGH)

प्रभारी वैज्ञानिक:
Scientist-in-charge:
(NIDHI SINGH)

एन पी एल आई आर - राष्ट्रीय भौतिक प्रयोगशाला
CSIR-NATIONAL PHYSICAL LABORATORY
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राष्ट्रीय मापिकी संस्थान (एनएमआई), सदस्य बीआईपीएस एवं हस्ताक्षरकर्ता सीआईपीएम - एमआरए)
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अंशांकन प्रमाण पत्र
CALIBRATION CERTIFICATE
A SET OF WEIGHTS

प्रमाण पत्र संख्या / Certificate No.
25060361/D1.01/C-051

दिनांक/ Date	अगले अंशांकन हेतु अनुशंसित तिथि Recommended date for next calibration	पृष्ठ / Page	पृष्ठों की संख्या / No. of Pa ges
22.08.2025	22.08.2028	2	2

7. Results :

Denomination	Mass Value (g)		Uncertainty (g)
10 kg	9999.985 2	*	± 0.005 0
5 kg	4999.992 9	*	± 0.002 5
2 kg	1999.996 7	*	± 0.001 0
.2 kg	1999.998 5		± 0.001 0
1 kg	999.999 67		± 0.000 50
500 g	499.999 67		± 0.000 25
200 g	200.000 04		± 0.000 10
. 200 g	199.999 88		± 0.000 10
100 g	99.999 95		± 0.000 05
50 g	50.000 02		± 0.000 03
20 g	20.000 020		± 0.000 025
. 20 g	20.000 017		± 0.000 025
10 g	10.000 037		± 0.000 020
5 g	4.999 969		± 0.000 016
2 g	2.000 007		± 0.000 012
. 2 g	2.000 003		± 0.000 012
1 g	1.000 000		± 0.000 010
500 mg	0.499 987		± 0.000 008
200 mg	0.200 004		± 0.000 006
. 200 mg	0.200 001		± 0.000 006
100 mg	0.100 010		± 0.000 005
50 mg	0.050 006		± 0.000 004
20 mg	0.020 001		± 0.000 003
. 20 mg	0.020 004		± 0.000 003
10 mg	0.010 004		± 0.000 003
5 mg	0.005 001		± 0.000 003
2 mg	0.002 002		± 0.000 003
. 2 mg	0.002 002		± 0.000 003
1 mg	0.001 002		± 0.000 003

The reported uncertainty is at coverage factor $k = 2$ which corresponds to a coverage probability of approximately 95 % for a normal distribution.

8. Date(s) of Calibration : 06.08.2025

9. Remarks :
1. Mass value(s) of the weight(s) except that(those) marked with asterisk(*) is(are) within the maximum errors permissible in E_2 accuracy class of weights as per OIML R 111-1 : 2004.
2. Dots(.) are used to distinguish the weights of same nominal value.

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